G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

Department of Computer Science and Engineering

B. Tech Minor in Data Science

III B. Tech. I Semester Academic Year 2026-27

Assignment-2

SUBJECT: Introduction to Data Science

SUBJECT CODE: U3111 **Max Marks: 5**

BRANCH: IT, ETM

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 - Creating

Answer all questions.

S. No	Question	CO	BT Level
1a)	What is a factor in R? How does it differ from a character vector?	CO3	L2
1b)	Explain how to iterate over a list in R using looping constructs with a suitable example.	CO3	L3
2a)	What are data frames in R? How they are different from vectors and lists?	CO3	L1
2b)	Create a data frame with columns: Name, Age, and Department. Add a new column Salary to the existing data frame. Add a new row representing a new employee.	CO3	L3
3a)	Explain the need for data reduction strategies in data science course.	CO4	L1
3b)	Analyse the usage of 'see' package. Refer the paper: Lüdecke, Daniel, et al. "see: An R package for visualizing statistical models." *Journal of Open Source Software* 6.64 (2021): 3393.	CO4	L4
4 a)	Write a program to find the GCD of given two number. Trace the program for three iterations.	CO3	L3
4 b)	Write a program to find the factorial of given number using recursion. Trace the program for three iterations.	CO3	L3
5a)	Explain the principle behind geometric projection-based visualization. What challenges arise when visualizing high dimensional data with geometric projection techniques?	CO4	L4
5b)	Write a program to sort the given numbers using quick sort. Trace the program for three iterations.	CO3	L3